

# A Probabilistic Analysis and Optimization of a Casino Game

Andrei Dragomir

December 2024

## Abstract

We construct a simple casino game and analyze it using probability theory, with a light measure-theoretic framing. After describing the game, we derive the distribution of the round on which a player loses, introduce a payout function, and choose the game's parameters so that the house keeps a small, controlled edge. We then play the game—both in a private test run and at a real casino night—and compare the resulting data against our theoretical predictions. We find that the test data agree closely with the theory, while the casino data, although unusually favourable to the house on the day, remain consistent with the model once sampling variability is taken into account.

## 1 Introduction

The game we propose is played on an  $n \times n$  array of tiles, of which  $m$  are *faulty* and the remaining  $n^2 - m$  are *safe*. The player pays  $x$  dollars to play and then turns over tiles one at a time. As long as the player keeps turning over safe tiles, the game continues. The moment the player turns over a faulty tile the game ends, and the player collects a payout determined by how many rounds they survived. Our goal is to choose  $n$ ,  $m$ ,  $x$ , and the payout function so that the game is fun to play yet profitable for the house.

Throughout we use a small amount of measure theory to keep the bookkeeping clean. A reader unfamiliar with measures may wish to read **Appendix A** first, where the few facts we need are collected. Our notation is summarized in **Appendix B**.

## 2 The Probability Model

### 2.1 Sample space and random variables

The natural sample space for this game is the set of rounds on which the player can lose. Since the player can fail as early as the first round and can survive at most all  $n^2 - m$  safe tiles before being forced onto a faulty one, we take

$$\Omega := \{1, 2, \dots, n^2 - m + 1\}.$$

An outcome  $\omega \in \Omega$  records the round on which the player loses. We let  $X: \Omega \rightarrow \mathbb{R}$  be the random variable giving this losing round, so that  $X(\omega) = \omega$ .

The player's winnings are determined by a *payout function*  $f: \Omega \rightarrow [0, \infty)$ , where  $f(\omega)$  is the amount paid out for losing on the  $\omega$ -th round. Because losing immediately (on the

first round) should earn nothing, we impose the constraint  $f(1) = 0$ . The winnings are then the random variable

$$W := f \circ X, \quad W(\omega) = f(\omega),$$

whose support (set of possible values) is the image  $f(\Omega)$ . We write the casino's profit as the random variable

$$\Pi := x - W, \quad \Pi(\omega) = x - f(\omega).$$

### 2.1.1 Two example payout functions

If the player earns a flat  $c$  dollars for each round survived, then

$$f(\omega) = c(\omega - 1), \quad \text{so } W \text{ takes the values } \{c(\omega - 1) : \omega \in \Omega\}.$$

If instead the player's winnings double each round, then

$$f(\omega) = \begin{cases} 2^{\omega-1} & \text{if } \omega \neq 1, \\ 0 & \text{if } \omega = 1, \end{cases} \quad \text{so } W \text{ takes the values } \{0\} \cup \{2^{\omega-1} : 2 \leq \omega \leq n^2 - m + 1\}.$$

Both of these are injective, so distinct rounds give distinct payouts. We choose the payout we actually use in **Section 3.3**; it will *not* be injective, and we deal with that case carefully in **Section 3.3.1**.

## 2.2 The distribution of the losing round

We now derive  $\mathbb{P}(X = \omega)$ , the probability of losing on the  $\omega$ -th round. Losing on the first round just means turning over a faulty tile first, so

$$\mathbb{P}(X = 1) = \frac{m}{n^2},$$

since there are  $m$  faulty tiles among  $n^2$  in total. Likewise the probability of starting with a safe tile is  $(n^2 - m)/n^2$ . More generally, suppose the player has already survived  $\omega$  safe picks. Then  $n^2 - m - \omega$  safe tiles remain among  $n^2 - \omega$  tiles, so the probability of surviving the next pick is

$$\frac{n^2 - m - \omega}{n^2 - \omega}.$$

Multiplying these conditional probabilities, the chance of surviving the first  $\omega$  picks is

$$\prod_{\alpha=0}^{\omega-1} \frac{n^2 - m - \alpha}{n^2 - \alpha}.$$

To lose on round  $\omega$ , the player must survive the first  $\omega - 1$  picks and then turn over a faulty tile, which (with  $\omega - 1$  tiles already removed) happens with probability  $m/(n^2 - \omega + 1)$ . Combining,

$$\mathbb{P}(X = \omega) = \frac{m}{n^2 - \omega + 1} \prod_{\alpha=0}^{\omega-2} \frac{n^2 - m - \alpha}{n^2 - \alpha}, \quad \omega \geq 2,$$

and  $\mathbb{P}(X = 1) = m/n^2$ , as above. (The general formula in fact reduces to  $m/n^2$  at  $\omega = 1$ , since the empty product equals 1.)

There is a cleaner way to write the survival probability. The number of ways to pick  $\omega$  safe tiles is  $\binom{n^2-m}{\omega}$  and the number of ways to pick any  $\omega$  tiles is  $\binom{n^2}{\omega}$ , so surviving  $\omega$  picks has probability  $\binom{n^2-m}{\omega}/\binom{n^2}{\omega}$ . Specializing to surviving the first  $\omega-1$  picks and then failing gives the compact formula we use from here on:

$$\mathbb{P}(X = \omega) = \frac{m}{n^2 - \omega + 1} \cdot \frac{\binom{n^2-m}{\omega-1}}{\binom{n^2}{\omega-1}}.$$

That the two expressions agree is the content of the following proposition.

**Proposition 1.** *For every integer  $k$  with  $0 \leq k \leq n^2 - m$ ,*

$$\prod_{\alpha=0}^{k-1} \frac{n^2 - m - \alpha}{n^2 - \alpha} = \frac{\binom{n^2-m}{k}}{\binom{n^2}{k}}.$$

*Proof.* Using  $\binom{N}{k} = N!/(k!(N-k)!)$  with  $N = n^2$  and  $N = n^2 - m$ ,

$$\frac{\binom{n^2-m}{k}}{\binom{n^2}{k}} = \frac{\frac{(n^2-m)!}{k!(n^2-m-k)!}}{\frac{(n^2)!}{k!(n^2-k)!}} = \frac{(n^2-m)!(n^2-k)!}{(n^2)!(n^2-m-k)!}.$$

Cancelling the factorials term by term leaves the falling products

$$\frac{(n^2-m)(n^2-m-1)\cdots(n^2-m-k+1)}{n^2(n^2-1)\cdots(n^2-k+1)} = \prod_{\alpha=0}^{k-1} \frac{n^2-m-\alpha}{n^2-\alpha}. \quad \square$$

## 2.3 Expected profit and the target margin

Let  $\mathbb{E}[W]$  denote the player's expected winnings and  $\mathbb{E}[\Pi]$  the casino's expected profit. Summing payout against probability,

$$\mathbb{E}[W] = \sum_{\omega \in \Omega} f(\omega) \mathbb{P}(X = \omega) = m \sum_{\omega \in \Omega} \frac{\binom{n^2-m}{\omega-1} f(\omega)}{(n^2 - \omega + 1) \binom{n^2}{\omega-1}}.$$

We want the house to keep between 0.10 and 0.25 dollars per dollar wagered, i.e.  $0.10x \leq \mathbb{E}[\Pi] \leq 0.25x$ . Since  $\mathbb{E}[\Pi] = x - \mathbb{E}[W]$ , this is equivalent to

$$0.75 \leq \frac{\mathbb{E}[W]}{x} \leq 0.90, \quad \text{that is,} \quad 0.75 \leq \frac{m}{x} \sum_{\omega \in \Omega} \frac{\binom{n^2-m}{\omega-1} f(\omega)}{(n^2 - \omega + 1) \binom{n^2}{\omega-1}} \leq 0.90.$$

This is the inequality we must satisfy when choosing parameters.

## 2.4 The law of $W$ as a measure

It is convenient to record the model as a probability space. We equip  $\Omega$  with the counting-style measure

$$\mu(\{\omega\}) := \mathbb{P}(X = \omega), \quad \omega \in \Omega,$$

which is a genuine probability measure because  $\{\{\omega\} : \omega \in \Omega\}$  partitions  $\Omega$  and

$$\mu(\Omega) = \sum_{\omega \in \Omega} \mu(\{\omega\}) = \sum_{\omega \in \Omega} \mathbb{P}(X = \omega) = 1.$$

Integration against  $\mu$  is then just summation, so for any  $a, b \in \mathbb{N}$ ,

$$\mathbb{P}(W \in [a, b]) = \int_{\{\omega : f(\omega) \in [a, b]\}} 1 \, d\mu = \sum_{\substack{\omega \in \Omega \\ f(\omega) \in [a, b]}} \mathbb{P}(X = \omega).$$

The advantage of this viewpoint is that the law of  $W$  is the *pushforward* of  $\mu$  along  $f$ : the probability of any payout is obtained by collecting all rounds that produce it. This is exactly what we need once  $f$  fails to be injective, which we take up in **Section 3.3.1**.

## 2.5 How likely is an observed sample?

*The symbols in this subsection are generic and are not tied to the specific game.*

Let  $\{X_i\}_{i=1}^n$  be independent and identically distributed (i.i.d.) copies of a random variable  $X$ , and let

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

be the sample mean. By linearity of expectation and  $\mathbb{E}[X_i] = \mathbb{E}[X]$ ,

$$\mathbb{E}[\bar{X}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mathbb{E}[X].$$

Using  $\text{Var}(cY) = c^2 \text{Var}(Y)$  together with independence (so the variance of the sum is the sum of the variances),

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n} \text{Var}(X).$$

Writing  $\sigma_X = \sqrt{\text{Var}(X)}$ , this says

$$\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}}, \quad \text{where} \quad \sigma_X = \sqrt{\sum_i (x_i - \mathbb{E}[X])^2 \mathbb{P}(X = x_i)}.$$

By the Central Limit Theorem, as  $n \rightarrow \infty$  the sample mean is approximately normal:

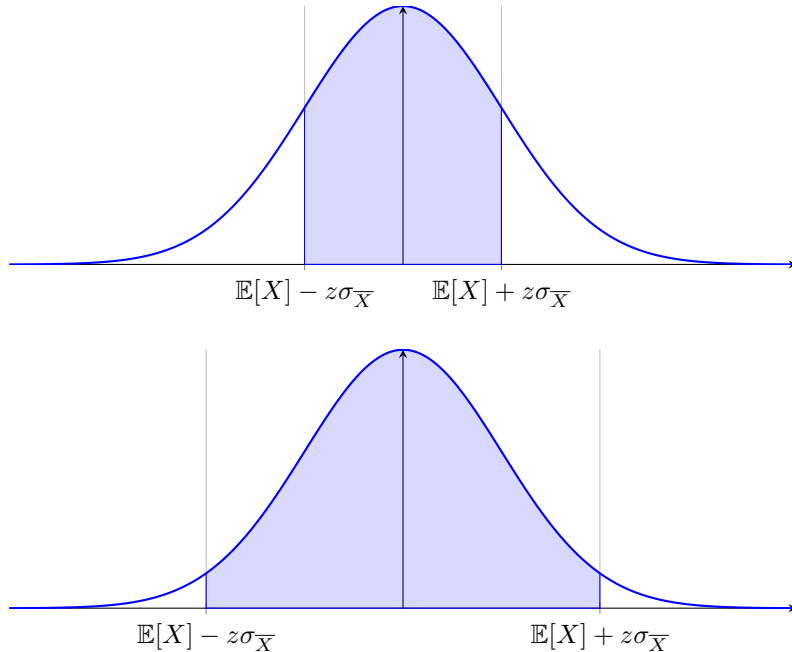
$$\bar{X} \approx \mathcal{N}(\mathbb{E}[X], \frac{1}{n} \text{Var}(X)).$$

(We need not worry about the sample size being too large, since in principle the game can be played arbitrarily many times.)

Suppose we observe a sample mean  $\bar{X} = k$ . To judge how surprising this is, we compute its  $z$ -score, which measures how many standard deviations it sits from the expected value:

$$z_k = \frac{k - \mathbb{E}[X]}{\sigma_X / \sqrt{n}}.$$

A small  $|z_k|$  places the observed mean near the centre of the normal curve, where the density is high, so such a sample is likely; a large  $|z_k|$  pushes it into the tails, where the density is low, so such a sample is unlikely. The two curves below illustrate this: the shaded interval  $\mathbb{E}[X] \pm z\sigma_{\bar{X}}$  is narrow for a small  $z$  (top) and wide for a large  $z$  (bottom).



Thus, by approximating the sampling distribution of the mean by a normal curve, we can take experimental data and decide whether the observed average is typical or surprising—which in turn tells us whether the game behaved as the model predicts.

### 3 Choosing the Game's Parameters

#### 3.1 Constraints

To keep the game realistic we impose, in addition to the margin constraint of **Section 2.3**:

$1 \leq x \leq 5$ ,  $x \in \mathbb{Z}$ . We want the game to be affordable.

$4 \leq n \leq 7$ ,  $n \in \mathbb{Z}$ . Too few tiles makes the game dull; too many makes it hard to build.

$m \approx \lceil n^2/5 \rceil$ ,  $m \in \mathbb{Z}$ , where  $\lceil \cdot \rceil$  rounds to the nearest integer. This keeps the early rounds easy (a roughly 80% chance of surviving the first round), so the game feels welcoming.

#### 3.2 Design goals for the payout

We tested linear, exponential, polynomial, and several hybrid payout functions; details are in **Appendix C**. Given the tight constraints, it was hard to find triples  $(n, m, x)$  that both satisfied the margin and kept the game enjoyable. Our guiding principles were:

- (i) the ratio  $n^2 : m$  should be small, so the game *looks* easier than it is and players rarely lose in the opening rounds;
- (ii) most players should break even or lose only a little, which sustains engagement; and
- (iii)  $f$  should be increasing in  $\omega$ , so surviving longer always pays more.

### 3.3 Constructing the payout function

We fix  $n = 4$ ,  $m = 2$ , and  $x = 3$ : this limits the materials needed, makes the game look trivial (only two faulty tiles), and keeps the entry price affordable. We then need an integer-valued  $f: \Omega \rightarrow [0, \infty)$  with  $f(1) = 0$ , increasing in  $\omega$ , but growing slowly enough that the expected payout lands in our target range. A suitable choice is

$$f(\omega) = \begin{cases} \lceil \frac{1}{2}(\omega - 1) \rceil & \text{if } 1 \leq \omega \leq 14, \\ 15 & \text{if } \omega = 15, \end{cases}$$

which we tabulate below.

Table 1: Values of the payout function  $f(\omega)$ .

|             |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
|-------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|
| $\omega$    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| $f(\omega)$ | 0 | 1 | 1 | 2 | 2 | 3 | 3 | 4 | 4 | 5  | 5  | 6  | 6  | 7  | 15 |

Using the formula of **Section 2.3**, this gives

$$\mathbb{E}[W] = \sum_{\omega \in \Omega} f(\omega) \mathbb{P}(X = \omega) \approx 2.633 \quad \implies \quad \mathbb{E}[\Pi] \approx 0.367.$$

Dividing by the entry price, the expected profit per dollar wagered is

$$\frac{\mathbb{E}[\Pi]}{x} = \frac{0.367}{3} \approx 0.122 \text{ dollars per dollar,}$$

comfortably inside the target band  $[0.10, 0.25]$ .

#### 3.3.1 Generalizing the law of $W$ for non-injective payouts

The payout  $f$  is plainly not injective: for instance  $f(3) = f(4) = 1$  (see Table 1). So the probability of winning a particular amount  $\xi$  pools several rounds together. For a payout value  $\xi$ , define the fibre

$$\mathcal{G}_\xi := f^{-1}(\xi) = \{ \omega \in \Omega : f(\omega) = \xi \},$$

the set of rounds that pay out  $\xi$ . By the pushforward description of **Section 2.4**,

$$\mathbb{P}(W = \xi) = \int_{\mathcal{G}_\xi} 1 d\mu = \sum_{\omega \in \mathcal{G}_\xi} \mathbb{P}(X = \omega).$$

For example,  $\mathbb{P}(W = 2) = \mathbb{P}(X = 4) + \mathbb{P}(X = 5)$ .

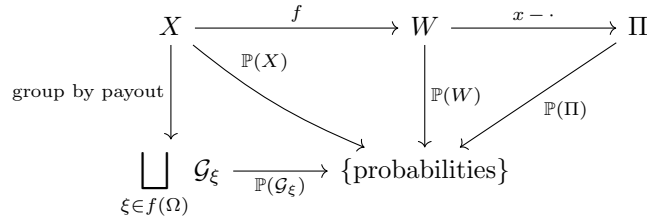
It is sometimes useful to single out one round in the fibre. Partitioning  $\mathcal{G}_\xi = \{\omega\} \cup (\mathcal{G}_\xi \setminus \{\omega\})$ , the probability of receiving the payout  $f(\omega)$  is

$$\mathbb{P}(W = f(\omega)) = \mathbb{P}(X = \omega) + \sum_{\omega' \in \mathcal{G}_\xi \setminus \{\omega\}} \mathbb{P}(X = \omega').$$

The first term is the chance of losing on round  $\omega$  specifically; the sum accounts for every other round that happens to pay the same amount. For our game this partition is more machinery than we need, but it is the right statement for a general payout.

## 4 A Summary Diagram

The relationships in the model are summarized by the following commutative diagram.



The losing round  $X$  determines the winnings  $W$  through the payout  $f$ , and  $W$  determines the house profit  $\Pi$  through  $x - \cdot$ . The rounds also partition by payout into the fibres  $\mathcal{G}_\xi$ ; collecting each fibre's round probabilities recovers the law of  $W$ . Every object maps to its set of probabilities under  $\mathbb{P}$ .

## 5 The Distribution

### 5.1 Outcomes

With the parameters and payout fixed, the supports of  $W$  and  $\Pi$  are

$$W \in \{0, 1, 2, 3, 4, 5, 6, 7, 15\} \quad \text{and} \quad \Pi \in \{-12, -4, -3, -2, -1, 0, 1, 2, 3\}.$$

### 5.2 Distribution table

Table 2 lists, for each round, the probability of losing on that round and the probability of the associated payout. Rounds that share a payout are grouped, illustrating the pooling of **Section 3.3.1**.

Table 2: Probability distribution.

| $\omega$ | $f(\omega)$ | $\mathbb{P}(X = \omega)$ | $\mathbb{P}(W = f(\omega))$ |
|----------|-------------|--------------------------|-----------------------------|
| 1        | 0           | 0.1250                   | 0.1250                      |
| 2        | 1           | 0.1167                   | 0.2250                      |
| 3        |             | 0.1083                   |                             |
| 4        | 2           | 0.1000                   | 0.1918                      |
| 5        |             | 0.0918                   |                             |
| 6        | 3           | 0.0833                   | 0.1583                      |
| 7        |             | 0.0750                   |                             |
| 8        | 4           | 0.0667                   | 0.1250                      |
| 9        |             | 0.0583                   |                             |
| 10       | 5           | 0.0500                   | 0.0917                      |
| 11       |             | 0.0417                   |                             |
| 12       | 6           | 0.0333                   | 0.0583                      |
| 13       |             | 0.0250                   |                             |
| 14       | 7           | 0.0167                   | 0.0167                      |
| 15       | 15          | 0.0083                   | 0.0083                      |

### 5.3 The distribution of $W$

Figure 1 plots the distinct payout values against their probabilities.

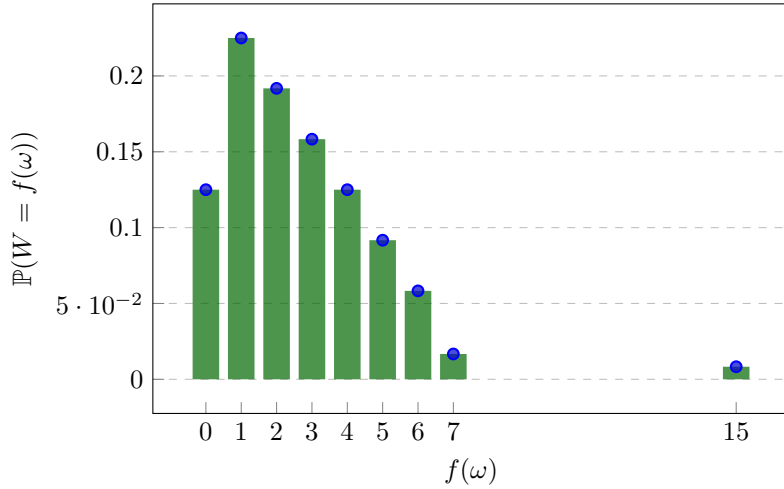


Figure 1:  $f(\omega)$  versus  $\mathbb{P}(W = f(\omega))$ .

The distribution is strongly skewed toward small payouts. Payouts near 0 or 1 are the most common, with  $\mathbb{P}(W = 1) \approx 0.225$  the largest. As  $\omega$  grows, surviving further becomes steadily less likely, so the probability mass thins out toward the higher payouts. Two values break the otherwise monotone trend:  $W = 0$  comes from the single round  $\omega = 1$ , whereas  $W = 1$  pools two rounds; and the jump from  $W = 7$  to  $W = 15$  is an artefact of the piecewise definition ( $f(14) = 7$ ,  $f(15) = 15$ ).



This shape is exactly what we want. Because there are only two faulty tiles, players regularly see at least a small reward, which keeps them interested and feeling that success is within reach. At the same time, large payouts are rare enough that the house holds its expected margin over many plays.

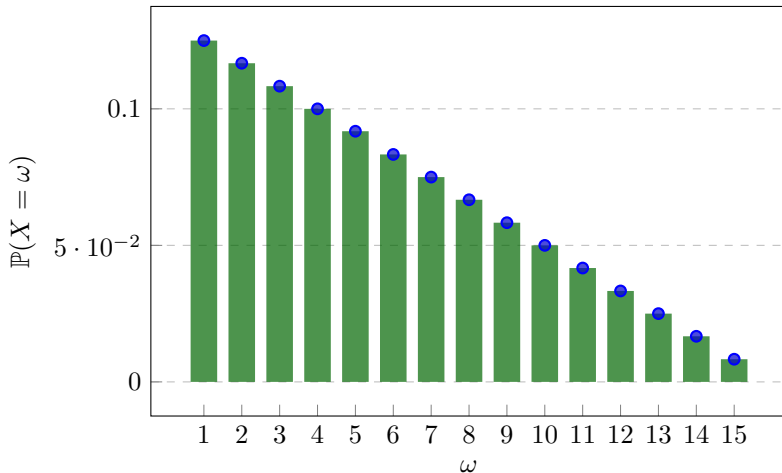


Figure 2:  $\omega$  versus  $\mathbb{P}(X = \omega)$ .

Notably,  $\mathbb{P}(X = \omega)$  is exactly linear in  $\omega$ : a short computation shows  $\mathbb{P}(X = \omega) = (16 - \omega)/120$  for our parameters. This monotone decline in the round probabilities is precisely what concentrates the winnings near small payouts. By rewarding the player often with small amounts while keeping large prizes rare, the design encourages repeat play and keeps the house margin in the intended range.

The remaining figures show, out of curiosity, how  $\mathbb{P}(X = \omega)$  changes as the number of faulty tiles  $m$  varies (with  $n = 4$  fixed).

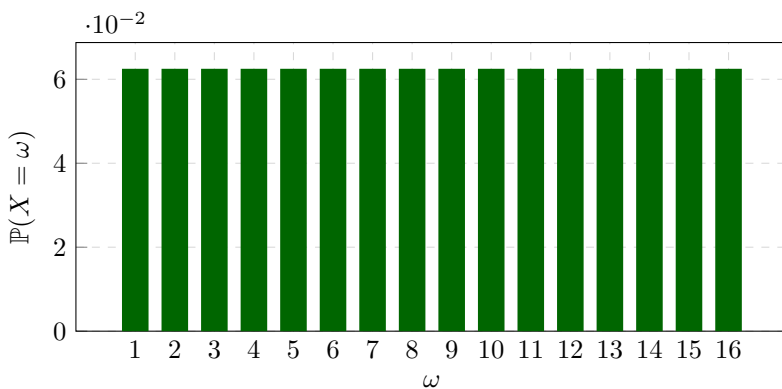


Figure 3:  $\omega$  versus  $\mathbb{P}(X = \omega)$  for  $m = 1, n = 4$ .

For  $m = 1$  the distribution is *uniform*, which at first seems paradoxical: as the player removes safe tiles, the chance of next hitting the single faulty tile should rise. Writing out

the round probabilities resolves it,

$$\underbrace{\frac{1}{16}}_{\omega=1} \underbrace{\frac{15}{16} \cdot \frac{1}{15}}_{\omega=2} \underbrace{\frac{15}{16} \cdot \frac{14}{15} \cdot \frac{1}{14}}_{\omega=3} \underbrace{\frac{15}{16} \cdot \frac{14}{15} \cdot \frac{13}{14} \cdot \frac{1}{13}}_{\omega=4} \dots$$

Every product telescopes to  $\frac{1}{16}$ : the rising conditional chance of hitting the faulty tile is exactly cancelled by the falling chance of surviving long enough to reach that round.

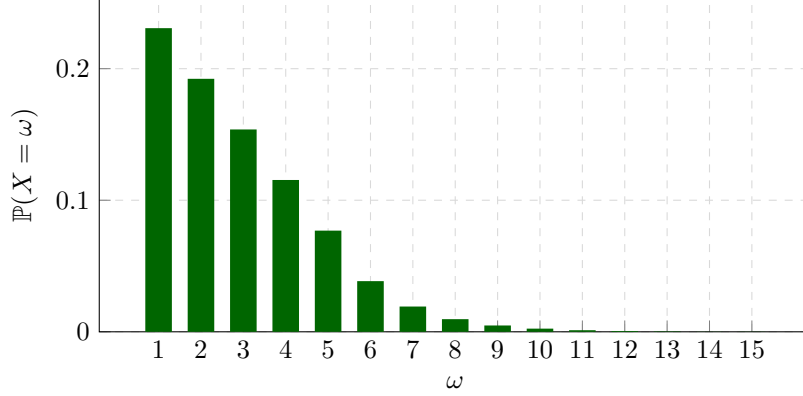


Figure 4:  $\omega$  versus  $\mathbb{P}(X = \omega)$  for  $m = 3, n = 4$ .

For  $m = 3$  the distribution skews toward earlier rounds, since a faulty tile is now more likely to appear soon.

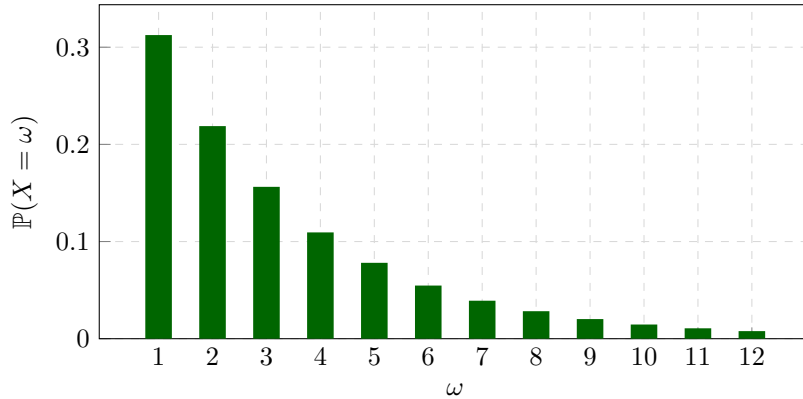


Figure 5:  $\omega$  versus  $\mathbb{P}(X = \omega)$  for  $m = 4, n = 4$ .

With  $m = 4$  the skew is sharper still, concentrating the mass in the first few rounds (note the different vertical scale from Figure 4).

## 6 The Game in Practice

Having fixed the model, we describe how the game is actually played. It uses a standard deck arranged as a  $4 \times 4$  grid of cards, of which two are designated “mines.” We shuffle a

sub-deck of 16 cards and lay them out at random, with black cards safe and the two red cards the mines. The player pays \$3 and flips cards one at a time until a red card appears. The payout for flipping the first red card on round  $\omega$  is:

$$\underbrace{1}_{\$0} \quad \underbrace{2 \ 3}_{\$1} \quad \underbrace{4 \ 5}_{\$2} \quad \underbrace{6 \ 7}_{\$3} \quad \underbrace{8 \ 9}_{\$4} \quad \underbrace{10 \ 11}_{\$5} \quad \underbrace{12 \ 13}_{\$6} \quad \underbrace{14}_{\$7} \quad \underbrace{15}_{\$15}$$

For example, flipping a red card on the 8th round wins the player \$4.

## 7 Experiment

To test the model we recorded the round each player failed on, from which all payouts and profits follow. We ran two experiments: a private test sample (us playing the game) and a live casino sample (the game run at a casino night). The raw data are in **Appendix D**.

### 7.1 Test sample

In the test run we collected  $n = 35$  plays and made a profit of \$15.

Table 3: Summary statistics for the test sample.

|                 | Mean   | Profit per dollar | Winnings per dollar |
|-----------------|--------|-------------------|---------------------|
| House profit    | 0.4286 | 0.1429            | –                   |
| Player winnings | 2.5714 | –                 | 0.8571              |

The observed means sit close to the theoretical  $\mathbb{E}[\Pi] \approx 0.367$  and  $\mathbb{E}[W] \approx 2.633$ . To quantify this, we approximate the sampling distribution of the mean winnings (for  $n = 35$ ) by  $\mathcal{N}(\mu = 2.633, \sigma_{\bar{X}} = 0.361)$ , using  $\sigma_W \approx 2.137$  from the distribution and  $\sigma_{\bar{X}} = \sigma_W / \sqrt{35}$ . The observed mean has  $z$ -score

$$z = \frac{2.5714 - 2.633}{0.361} \approx -0.17,$$

so the sample mean lies less than a fifth of a standard deviation below the expected winnings—squarely in the high-density centre of the curve, and exactly what we would expect from a faithful model. To corroborate this, Figure 6 shows the empirical payout frequencies, which we compare with the theoretical shape of **Section 5.3**.

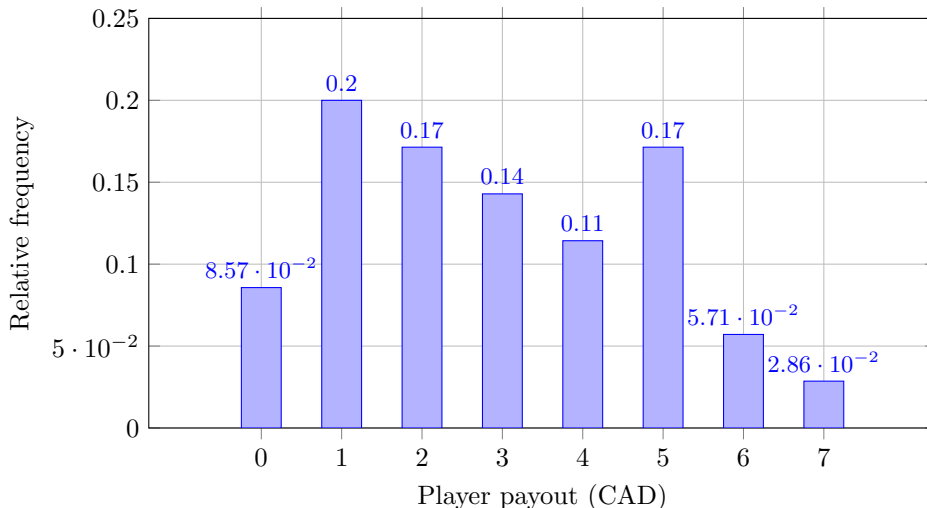


Figure 6: Test sample: player payout versus relative frequency.

The empirical histogram tracks the theoretical shape closely. The one conspicuous departure is the spike at a payout of \$5, which is well within ordinary sampling variation. Between the small  $z$ -score and the matching shape, the test sample behaves just as the model predicts, which gives us confidence that the game was set up and recorded correctly.

## 7.2 Casino sample

At the casino night we collected  $n = 49$  plays and made a profit of \$44.

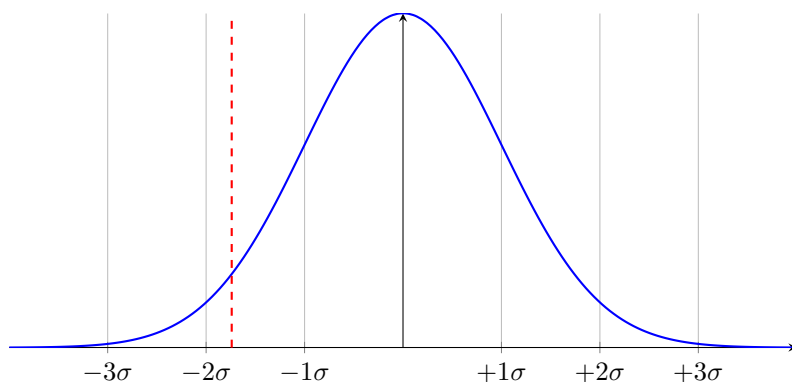
Table 4: Summary statistics for the casino sample.

|                 | Mean  | Profit per dollar | Winnings per dollar |
|-----------------|-------|-------------------|---------------------|
| House profit    | 0.898 | 0.299             | —                   |
| Player winnings | 2.102 | —                 | 0.701               |

Here the means depart noticeably from the theoretical values: the house took about \$0.30 per dollar wagered, well above the target band. Approximating the sampling distribution of the mean winnings (for  $n = 49$ ) by  $\mathcal{N}(\mu = 2.633, \sigma_{\bar{X}} = 0.305)$  gives

$$z = \frac{2.102 - 2.633}{0.305} \approx -1.74,$$

so the observed mean sits about 1.74 standard deviations below expectation—out in the left tail, as the figure below illustrates.



A mean this far into the tail is uncommon, but it worked in the house's favour: the casino took considerably more than its expected margin that day. Figure 7 shows the empirical payouts.

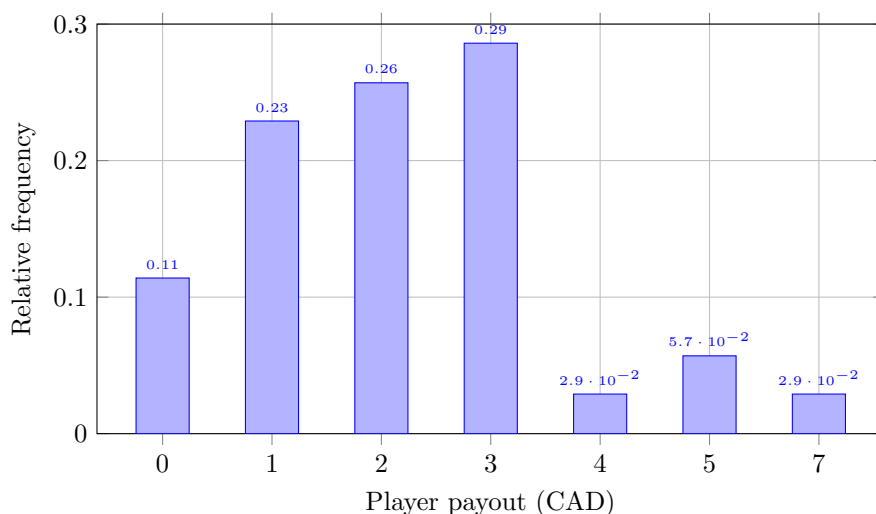


Figure 7: Casino sample: player payout versus relative frequency.

The empirical distribution differs more from the theory than the test sample did, though such deviation is plausible given sampling variability, and it retains the expected left-skew. There is a pronounced peak at a payout of \$3 (relative frequency  $\approx 0.29$ ) followed by a sharp drop for larger payouts, meaning most players broke even or lost: players failed to come out ahead about 60% of the time. Given the large negative  $z$ -score and this pronounced skew, the sample is on the unlikely side—it appears we simply had an unusually favourable day.

## 8 Discussion

The casino night was a success. We took a total profit of \$44, about \$0.90 per game, or roughly \$0.30 per dollar wagered—comfortably above our expected margin. The headline

reason is that players failed to come out ahead about 60% of the time. Yet despite usually losing, many players returned for repeated games, which we attribute to the design.

With only two faulty tiles, players read the game as a trivial challenge (**Sections 3.1 and 3.2**). Even after early losses they kept playing, seemingly chalking failures up to bad luck rather than the game’s odds. Letting players choose their own tiles created an illusion of skill that further encouraged replays (**Section 3.2**), an effect that persisted through repeated losses—some players continued even after winning nothing in the first round. The combination of perceived control and apparent ease kept engagement high, in line with our goal of balancing enjoyment against profitability.

## 8.1 Future improvements

The design worked well, so our proposed improvements keep the same structure.

| Improvement         | Explanation                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Add betting         | Letting players bet raises the potential winnings, which can make the game more engaging.                                                |
| Better presentation | A flashier visual design would draw players in and increase turnout, and hence profit.                                                   |
| Add a jackpot       | A progressive jackpot fed by a slice of each entry fee gives players an extra incentive to keep playing for a chance at a growing prize. |

Table 5: Proposed improvements.

## 9 Conclusion

We constructed a casino game designed to balance profitability against player engagement, and analyzed it using probability theory with a measure-theoretic framing. We derived the law of the losing round, built a payout function keeping the house margin inside a chosen band, and validated the model against two experiments. The test sample matched the theory closely, while the casino sample—though favourable to the house on the day—remained consistent with the model once sampling variability was accounted for. Together these confirm that the game performs as designed.

## A Appendix: What Is a Measure?

A **measure** assigns a non-negative size to subsets of a set, generalizing length, area, and volume.

Let  $X$  be a set and let  $\mathcal{A}$  be a  $\sigma$ -algebra of subsets of  $X$ . A measure is a function  $\mu: \mathcal{A} \rightarrow [0, \infty]$  satisfying:

1.  $\mu(A) \geq 0$  for all  $A \in \mathcal{A}$ ;
2.  $\mu(\emptyset) = 0$ ;
3. ( $\sigma$ -additivity) for any countable collection of disjoint sets  $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ ,

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

### A.1 Examples

**Length.** On  $\mathbb{R}$ , the length of  $[a, b]$  is  $\mu([a, b]) = b - a$ .

**Area.** On  $\mathbb{R}^2$ , the area of an  $l \times w$  rectangle is  $lw$ .

**Hypervolume.** On  $\mathbb{R}^k$ , the volume of an  $x_1 \times \cdots \times x_k$  box is  $\prod_{j=1}^k x_j$ .

**Probability.** A measure assigning probabilities to events. For a fair die,  $\mu(\{X = 1\}) = \frac{1}{6}$ .

### A.2 Integration against a measure

The integral of  $f$  with respect to  $\mu$ , written  $\int f d\mu$ , generalizes both summation and ordinary integration. When  $\mu$  is Lebesgue measure it recovers the usual integral for well-behaved  $f$ ; when  $\mu$  is a probability measure it computes the expected value of  $f$ .

### A.3 The discrete case

For a discrete random variable  $X$  taking values  $\{x_1, x_2, \dots\}$  with probabilities  $\{p_1, p_2, \dots\}$ , set  $\mu(\{x_i\}) = p_i$ . Then

$$\int f(X) d\mu = \sum_i f(x_i) p_i,$$

the expected value of  $f(X)$ . (For a continuous variable with density  $p$ , the integral becomes  $\int_{\mathbb{R}} f(x)p(x) dx$ .) In particular, with  $f \equiv 1$  on an interval,

$$\mathbb{P}(X \in [a, b]) = \int_{[a, b]} d\mu = \sum_{x \in [a, b]} \mathbb{P}(X = x).$$

In words: integrating against this measure is just a tidy way of writing  $\mathbb{P}(X \in [a, b])$ . This is the only kind of integral the paper uses.

## B Appendix: Notation and Glossary

$n, m, x$ : the grid side length, the number of faulty tiles, and the entry price; here  $n = 4$ ,  $m = 2$ ,  $x = 3$ .

$\Omega, \omega$ : the sample space  $\{1, \dots, n^2 - m + 1\}$  of possible losing rounds, and a generic outcome  $\omega \in \Omega$ .

$X, W, \Pi$ : the losing round, the player's winnings  $W = f \circ X$ , and the house profit  $\Pi = x - W$ .

$f$ : the payout function,  $f(\omega)$  being the payout for losing on round  $\omega$ .

$\mathcal{G}_\xi$ : the fibre  $f^{-1}(\xi)$ , i.e. all rounds paying out  $\xi$ .

**Summation**  $\sum$ :  $\sum_{n \in \mathcal{N}} a_n$  adds the terms  $a_n$  as  $n$  ranges over  $\mathcal{N}$ ; e.g.  $\sum_{n \in \{1,2\}} n = 1 + 2$ .

**Intervals**:  $[a, b] := \{k \in \mathbb{Z} : a \leq k \leq b\}$ , used to reduce clutter.

**Injective function**:  $f: A \rightarrow B$  with  $f(a_1) = f(a_2) \iff a_1 = a_2$ ; no two inputs share an image.

**Set notation**:  $\{x_i\}_{i=1}^k$  denotes the set  $\{x_1, \dots, x_k\}$ .

**Set difference**:  $A \setminus B$  is the set of elements in  $A$  but not in  $B$ .

**Increasing function**:  $f$  with  $a < b \iff f(a) < f(b)$ , i.e. strictly increasing.

**Union / intersection**:  $\bigcup_{i \in I} A_i$  collects elements in some  $A_i$ ;  $\bigcap_{i \in I} A_i$  collects elements in every  $A_i$ .

**Floor / ceiling**:  $\lfloor x \rfloor$  is the greatest integer  $\leq x$ ;  $\lceil x \rceil$  is the least integer  $\geq x$ .

## C Appendix: Payout Functions We Tested

Before settling on the payout of **Section 3.3** we tried linear, exponential, polynomial, and hybrid functions. The recurring problem with the standard families was that the player's expected winnings overshoot the margin constraints of **Sections 2.3 and 3.1**; the recurring problem with the hybrids was that players found it hard to break even. Two representative families are below.

### C.1 Dampened, capped Fibonacci payout

$$f(\omega) = \begin{cases} \lceil \lambda F_\omega \rceil & \text{if } \omega \leq k, \\ \min\{\omega, M\} & \text{if } \omega > k, \end{cases}$$

where  $F_\omega$  is the Fibonacci sequence. The table reports the house profit per dollar wagered.



| $\lambda$ | $M$ | $k$ | $n$ | $m$ | $x$ | House profit per dollar | $\mathbb{E}[W]$ |
|-----------|-----|-----|-----|-----|-----|-------------------------|-----------------|
| 0.3       | 30  | 8   | 4   | 3   | 3   | 0.1661                  | 2.5018          |
| 0.3       | 30  | 8   | 5   | 5   | 4   | 0.2370                  | 3.0520          |
| 0.3       | 30  | 10  | 4   | 3   | 3   | 0.1661                  | 2.5018          |
| 0.3       | 30  | 10  | 5   | 5   | 4   | 0.2370                  | 3.0520          |
| 0.3       | 30  | 12  | 4   | 3   | 3   | 0.1196                  | 2.6411          |
| 0.3       | 30  | 12  | 5   | 5   | 4   | 0.1933                  | 3.2270          |
| 0.3       | 70  | 8   | 5   | 5   | 5   | 0.1628                  | 4.1858          |
| 0.3       | 70  | 8   | 5   | 6   | 3   | 0.2095                  | 2.3714          |
| 0.3       | 70  | 10  | 5   | 5   | 5   | 0.1628                  | 4.1858          |
| 0.3       | 70  | 10  | 5   | 6   | 3   | 0.2095                  | 2.3714          |
| 0.3       | 70  | 12  | 5   | 5   | 5   | 0.1628                  | 4.1858          |
| 0.3       | 70  | 12  | 5   | 6   | 3   | 0.2095                  | 2.3714          |
| 0.5       | 30  | 8   | 4   | 3   | 5   | 0.1921                  | 4.0393          |
| 0.5       | 30  | 8   | 5   | 5   | 5   | 0.1022                  | 4.4892          |
| 0.5       | 50  | 10  | 4   | 4   | 3   | 0.2390                  | 2.2830          |
| 0.5       | 50  | 10  | 5   | 6   | 4   | 0.1092                  | 3.5630          |

Table 6: Results for the hybrid Fibonacci payout.

The drawback is that the payout caps out quickly, so beyond a certain round players gain nothing by continuing, which discourages play.

## C.2 Logistic payout

A logistic payout grows fast in the early rounds (giving the player hope of profit) and then flattens out:

$$f(\omega) = \left\lfloor \frac{M}{1 + e^{-k(\omega-t)}} \right\rfloor.$$

| $M$ | $k$ | $t$ | $n$ | $m$ | $x$ | House profit per dollar | $\mathbb{E}[W]$ |
|-----|-----|-----|-----|-----|-----|-------------------------|-----------------|
| 20  | 0.3 | 7   | 4   | 6   | 5   | 0.1024                  | 4.4879          |
| 20  | 0.3 | 8   | 4   | 5   | 5   | 0.2215                  | 3.8924          |
| 20  | 0.3 | 8   | 4   | 6   | 4   | 0.1365                  | 3.4539          |
| 20  | 0.3 | 9   | 4   | 5   | 4   | 0.2068                  | 3.1728          |
| 20  | 0.3 | 9   | 5   | 6   | 5   | 0.1827                  | 4.0867          |
| 20  | 0.3 | 10  | 4   | 5   | 3   | 0.2050                  | 2.3851          |
| 20  | 0.3 | 10  | 5   | 5   | 5   | 0.2295                  | 3.8526          |
| 20  | 0.3 | 10  | 5   | 6   | 4   | 0.1923                  | 3.2309          |
| 20  | 0.7 | 9   | 6   | 6   | 5   | 0.2064                  | 3.9680          |

Table 7: Results for the logistic payout.

This family failed because the floor function sent many early rounds to a payout of zero, which discourages players.

## D Appendix: Raw Experimental Data

Table 8: Test sample: player payout versus house profit.

| Game | Player payout (CAD) | House profit (CAD) |
|------|---------------------|--------------------|
| 1    | 1                   | 2                  |
| 2    | 0                   | 3                  |
| 3    | 5                   | -2                 |
| 4    | 4                   | -1                 |
| 5    | 2                   | 1                  |
| 6    | 2                   | 1                  |
| 7    | 3                   | 0                  |
| 8    | 1                   | 2                  |
| 9    | 6                   | -3                 |
| 10   | 5                   | -2                 |
| 11   | 2                   | 1                  |
| 12   | 5                   | -2                 |
| 13   | 5                   | -2                 |
| 14   | 2                   | 1                  |
| 15   | 1                   | 2                  |
| 16   | 2                   | 1                  |
| 17   | 3                   | 0                  |
| 18   | 1                   | 2                  |
| 19   | 5                   | -2                 |
| 20   | 7                   | -4                 |
| 21   | 1                   | 2                  |
| 22   | 2                   | 1                  |
| 23   | 4                   | -1                 |
| 24   | 4                   | -1                 |
| 25   | 1                   | 2                  |
| 26   | 0                   | 3                  |
| 27   | 0                   | 3                  |
| 28   | 1                   | 2                  |
| 29   | 4                   | -1                 |
| 30   | 4                   | -1                 |
| 31   | 0                   | 3                  |
| 32   | 2                   | 1                  |
| 33   | 1                   | 2                  |
| 34   | 1                   | 2                  |
| 35   | 3                   | 0                  |

Table 9: Casino sample: player payouts and house profits.

| Game | Player payout (CAD) | House profit (CAD) |
|------|---------------------|--------------------|
| 1    | 3                   | 0                  |
| 2    | 2                   | 1                  |
| 3    | 3                   | 0                  |
| 4    | 1                   | 2                  |
| 5    | 5                   | -2                 |
| 6    | 3                   | 0                  |
| 7    | 3                   | 0                  |
| 8    | 2                   | 1                  |
| 9    | 3                   | 0                  |
| 10   | 2                   | 1                  |
| 11   | 2                   | 1                  |
| 12   | 1                   | 2                  |
| 13   | 2                   | 1                  |
| 14   | 3                   | 0                  |
| 15   | 3                   | 0                  |
| 16   | 7                   | -4                 |
| 17   | 1                   | 2                  |
| 18   | 0                   | 3                  |
| 19   | 3                   | 0                  |
| 20   | 0                   | 3                  |
| 21   | 0                   | 3                  |
| 22   | 3                   | 0                  |
| 23   | 1                   | 2                  |
| 24   | 5                   | -2                 |
| 25   | 5                   | -2                 |
| 26   | 3                   | 0                  |
| 27   | 1                   | 2                  |
| 28   | 2                   | 1                  |
| 29   | 2                   | 1                  |
| 30   | 1                   | 2                  |
| 31   | 2                   | 1                  |
| 32   | 2                   | 1                  |
| 33   | 2                   | 1                  |
| 34   | 4                   | -1                 |
| 35   | 1                   | 2                  |
| 36   | 2                   | 1                  |
| 37   | 0                   | 3                  |
| 38   | 3                   | 0                  |
| 39   | 0                   | 3                  |
| 40   | 2                   | 1                  |
| 41   | 2                   | 1                  |

Table 10: Casino sample (continued).

| <b>Game</b> | <b>Player payout (CAD)</b> | <b>House profit (CAD)</b> |
|-------------|----------------------------|---------------------------|
| 42          | 1                          | 2                         |
| 43          | 3                          | 0                         |
| 44          | 3                          | 0                         |
| 45          | 1                          | 2                         |
| 46          | 2                          | 1                         |
| 47          | 3                          | 0                         |
| 48          | 0                          | 3                         |